LUMYVUE - system report

AI-automated facial skin screening system for acne, hyperpigmentation and eczema. Status as of 7 November 2026, written for the thesis defense (16 November 2026). Figures come from the project's saved evaluation files; limits are stated where they apply.

1. Summary

LUMYVUE is a complete screening workflow: a participant is registered and gives consent, three facial photographs are taken, an AI model reads each photograph, the result is shown in clinician-friendly language, a dermatologist reviews it, and every step is stored, filed and printable. The software side is complete, consistent and tested. The scientific evidence is moderate and unfinished: the AI is a useful screening aid, not a validated diagnostic tool.

Area	Status	Honest assessment
Workflow and screens	Complete	Registration, consent, live three-view capture, results, dermatologist review, receipt, A4 report, records and database screens. Consistent wording and layout.
AI result presentation	Complete	One outcome per scan (Condition detected / Needs review / No condition detected), per-condition result, certainty, facial regions, reasons for review, not a diagnosis.
Verification and records	Complete	Pending / Verified / Requires further review follows the dermatologist's decision; folders, database and audit log stay in step.
Software quality	Strong	103 automated tests; the application reproduces the reference model computation to about one part in a million.
AI accuracy	Moderate	On 132 held-out photographs: macro-F1 0.74; acne 0.87, hyperpigmentation 0.71, eczema 0.63. Good enough to screen and refer, not to diagnose.
Independent test	Not done	The locked test split has no verified labels. No result on fully unseen data exists.
Dermatologist validation	Not done	The review workflow is built and records agreement, but no systematic validation has been run.
Live camera and printer	Built, not rehearsed	Live reading, automatic capture and thermal receipt exist; end-to-end behaviour on the demonstration hardware still has to be rehearsed.

Bottom line: as a working system it is strong and the safety design (it defers to a dermatologist instead of guessing) is a real strength. As evidence, claim a screening aid with moderate measured accuracy and clearly stated limits.

2. How the system works

2.1 Workflow

Step	What happens	Where it is stored
1. Register	Patient's chart (personal data, present illness, medical history, optional examination). Patient ID PS-0001... is created automatically.	Participants, PatientCharts
2. Consent	Research and facial-image consent; scanning stays blocked until the required items are ticked. Each decision is time-stamped.	Consents
3. Capture	Live camera with a face guide. Front, left and right photographs, taken automatically once the face is steady, or by button or upload.	Patient Records\Pending verification\PatientID\date\scan
4. AI reading	Each photograph is analysed (about 2 s on the PC's CPU). A sealed result file is saved with the photograph.	v6_assessment.json (SHA-256 sealed); AIAssessments, AIFindings, AIMarkedAreas
5. Results	Outcome, findings table, reasons for review, photograph with marked areas, verification status.	Read from saved data; nothing is recomputed
6. Review	A dermatologist records their own assessment and agreement with the AI. Verified, Requires further review, or a draft.	DermatologistValidations; scan status; folder moves
7. Outputs	Thermal receipt (preliminary and final), A4 research report, patient record, research dashboard.	Receipts, print attempts, report from stored data

2.2 The AI model, in plain terms

- **Two readers of the same photograph.** A whole-face reader (a MobileNetV2 neural network) judges the overall appearance. Three specialist lesion detectors (YOLOv8, one each for acne, hyperpigmentation and eczema) look for individual spots and draw boxes.
- **One trained combiner.** A small neural network learns how much to trust each reader and outputs one probability per condition. Nothing is voted or averaged by hand.
- **Calibrated and thresholded.** Probabilities are calibrated so that they mean what they say, then compared with a per-condition decision limit (acne 0.42, hyperpigmentation 0.37, eczema 0.27).
- **Several conditions per photo.** A photograph can show more than one condition; 'normal' is not predicted but derived: nothing detected, all detectors working, no warning raised, quality checked.
- **Safety checks.** A result is sent to review when the two readers disagree, when a detector marked an area the combiner did not confirm, when the photograph looks unlike the training photographs, or when photo quality is poor.
- **Hyperpigmentation is detector-led.** An area marked by the hyperpigmentation detector is reported as Detected, labelled 'found by the lesion detector' with at most moderate certainty. This was chosen because the detector finds most hyperpigmentation photographs the combiner misses; the cost is more false alarms (section 4.3).

2.3 What the clinician sees

- **Outcome banner:** Condition detected, Needs review, No condition detected, Retake photograph, or Analysis not completed.
- **Findings table:** per condition - result (Detected, Possible, Inconclusive, Not detected), certainty (high, moderate, low), and the facial regions (forehead, right or left cheek, nose, mouth / chin) where the detectors marked areas.
- **Why review is needed:** plain-language reasons, grouped so the same reason is not repeated.
- **Marked areas:** the photograph with detector boxes, stated as possible locations, not a lesion count and not a diagnosis.
- **Technical details** (collapsed, for researchers): the calibrated probabilities against the decision limits.
- Every screen, the receipt and the A4 report carry the same statement: AI-assisted screening result, not a diagnosis, not clinically validated, must be confirmed by a dermatologist.

3. Data and training

Split	Photographs	Fully verified	Present (acne / hyperpigmentation / eczema)	Purpose
Training	5,873	889	176 / 261 / 243	Trains the whole-face network
Validation	979	591	173 / 233 / 55	Chooses models; the fair evaluation set is drawn from it
Fusion	1,469	347	111 / 124 / 33	Trains the combiner; calibration and decision limits
Test (locked)	1,468	0	0 / 0 / 0	Never opened: no verified labels yet

- **Splits are grouped** so identical or near-identical photographs never sit on both sides of a split.
- **Folder names are only weak labels** in training; evaluation uses verified labels only, and unknown labels are excluded rather than counted as negative.
- **The verified sets are small** (889 / 591 / 347 photographs; 18 eczema photographs in the fair set). Every accuracy figure below carries wide uncertainty.
- A re-review of 239 suspicious photographs changed only 12 labels, so label noise is not believed to be the main limit.

4. Measured performance

4.1 Held-out ('fair') evaluation set

132 fully verified photographs that the earlier model generation had never trained on: 47 with hyperpigmentation, 26 with no condition, 18 with eczema. Decisions use the deployed decision limits.

Measure	Result
Macro-F1 (average over three conditions)	0.737
F1 - acne / hyperpigmentation / eczema	0.870 / 0.711 / 0.632
Hyperpigmentation photographs for which no condition was named	10 of 47
Clear photographs wrongly flagged for hyperpigmentation	2 of 26

These numbers describe the combined AI before the detector-led hyperpigmentation rule and before the clinical wording was added. They are development-quality figures: the data were used to choose between candidate models, so they may be slightly optimistic.

4.2 Development evaluation (out-of-fold, 347 verified photographs)

Condition	Precision	Recall	F1	Average precision
Acne	0.876	0.892	0.884	0.950
Hyperpigmentation	0.918	0.903	0.911	0.952
Eczema	0.735	0.758	0.746	0.711

Overall exact-match accuracy 0.862 and macro-F1 0.846. This set was also used to calibrate and choose decision limits, so it is optimistic; the held-out set above is the fairer guide.

4.3 The hyperpigmentation rule: benefit and cost

- **Benefit:** the hyperpigmentation detector finds about 91% of hyperpigmentation photographs on the held-out set, against roughly 57% recall for the combined model alone.
- **Cost:** the same detectors flag about 27% of clear photographs. On 13 reference photographs the rule recovered 2 real hyperpigmentation cases and also marked hyperpigmentation on 1 clear face and 2 eczema photographs.
- **Mitigation in the product:** such findings are labelled 'found by the lesion detector', carry low or moderate certainty only, and always go to dermatologist review.

- **Not yet measured:** the false-alarm rate of the final application behaviour (combined model plus this rule) on a labelled set of clear faces. This is the single most useful number still missing.

4.4 What the stored test scans look like

The database holds 52 scans taken during development (uploads and live captures, many deliberately difficult). After analysis: about 62% 'Needs review', 35% 'Retake photograph' (mostly because the photograph failed the blur rule), 2 'Condition detected' and 1 'No condition detected'. These are development photographs, not a representative sample, but they show that the system is cautious: it prefers to defer to a dermatologist or ask for a retake.

5. Records, storage and database

- **Organised storage.** Every scan has one folder under Patient Records: Pending verification, Verified, Requires further review, or Unassigned scans, then PatientID, date and scan. The folder moves automatically when the verification status changes, and every stored path is rewritten in the same database transaction (tested: all 24 stored paths remained valid after each change).
- **Verification status.** Every new scan starts as Pending verification. A completed dermatologist review makes it Verified; a 'further review' outcome keeps it open; a draft changes nothing. A manual change needs a reason and is logged.
- **Database.** The existing database was checked (integrity, relationships, duplicates, missing files) and found healthy. It was upgraded additively with an automatic backup first: new tables for each photograph's AI result, per-condition findings and marked areas, plus a verification status and indexes. Nothing was deleted; the earlier model's stored values are kept for audit but no longer shown.
- **In-app database screen** ('Records & database'): patients, scans, AI findings, verification records and an activity log, each with search, filters and a details panel. It can register or edit patients, mark a patient withdrawn, change a scan's verification status, link an unassigned scan to a patient, open results, reviews and folders, back up the database and run an integrity check - without SQL or an external tool.
- **Audit trail.** Registration changes, scan saves, status changes, folder moves, AI results and backups are all recorded with who and when.
- **Reviews are append-only** and never alter the AI result; the sealed result file can be checked against its hash.

6. Quality assurance

- **103 automated tests** (all passing) cover the decision logic, the clinical reading rules, storage, receipts, reports and the model computation.
- **Faithfulness to the research code.** On 13 real photographs the application's detector boxes match the reference to 0.0001 pixel, the fusion output to about two parts in a million, with no decision flipped.
- **Integrity checks.** Each model file's hash is verified when the application starts; a mismatch stops the AI instead of giving unverified results.
- **Tested on a copy of the real data** (never the live database): migration of all 52 scans, the full review workflow, a new upload through the real scan pipeline, and rendered screenshots of the main screens.
- **Not covered by automation:** the live camera, the touch keyboard and the thermal printer.

7. Limitations and risks (read before the defense)

Limitation	Why it matters	How to state it
No independent test data	No result exists on photographs that played no part in building the system.	Say the locked test split awaits verified labels; the held-out set was used for model choice.
No dermatologist validation	Accuracy was measured against research labels, not a clinical reference standard.	Present the review workflow as the means by which this will be collected.
Small verified sets	Estimates are wide; eczema rests on 18 examples.	Give counts with every figure; avoid precise-sounding percentages for eczema.
Eczema is weakest	F1 0.63; fewest training examples.	Call eczema a pilot condition.
Hyperpigmentation false alarms	Detector-led rule marks some clear skin.	Explain the trade-off: fewer misses, more flags, always reviewed.
Many results need review or retake	A panel may read this as indecision.	It is deliberate caution; a missed condition is worse than a referral.
Skin-tone fairness not evaluated	Performance may differ across skin tones.	State as future work.

Limitation	Why it matters	How to state it
Photo-quality and 'unlike training' thresholds are provisional	Not validated against clinicians.	State as provisional.
Live camera and printer not rehearsed	Most likely source of a demonstration failure.	Rehearse twice on the real hardware and room.

8. Questions the panel is likely to ask

How accurate is it?

On 132 held-out photographs the macro-F1 was 0.74 (acne 0.87, hyperpigmentation 0.71, eczema 0.63). That is moderate: suitable for screening and referral, not for diagnosis. Eczema rests on 18 examples.

Was it tested on real patients?

Not yet in a clinical sense. The evaluation used verified photographs from our dataset. The system records dermatologist reviews and agreement so that clinical validation can follow.

Why does it so often say 'needs review'?

It is designed to defer when its two readers disagree, when it is unsure, or when photo quality is poor. A wrong confident answer is more harmful than a referral.

Does it diagnose?

No. Every screen, receipt and report says it is an AI-assisted screening result that a dermatologist must confirm.

Why two AI components?

The whole-face network sees overall appearance; the lesion detectors find individual spots and give locations. A trained combiner learns when to trust each, and the clinician sees one result with reasons.

What about hyperpigmentation false alarms?

The detector can mark clear skin. Those findings are labelled as detector-found, carry limited certainty, and always go to review. We have not yet measured the final false-alarm rate on a labelled clear-skin set; that is the next measurement.

How is patient data protected and traceable?

Results are sealed with a hash, every change is audited, scans are filed by verification status, the database is backed up before upgrades, and withdrawn patients are excluded from statistics.

What are the next steps?

Collect dermatologist reviews through the built-in workflow, evaluate on an independent labelled test set, measure false alarms on clear skin, evaluate across skin tones, and refine the decision limits.

9. Plan for the remaining days

Priority	Task	Why
1	Rehearse the full demonstration on the real camera, printer and room, twice; fix what breaks.	Most likely cause of a failed demonstration.
2	Measure the false-alarm rate of the final behaviour on labelled clear photographs and the recall on labelled hyperpigmentation photographs; add one honest table to the thesis.	The number the panel is most likely to ask for.
3	Collect as many dermatologist reviews as possible through the app.	Even a small number gives an AI-versus-dermatologist agreement figure.
4	Prepare short written answers to the questions in section 8.	Honest, rehearsed answers read as competence.
5	Freeze the build a few days before the defense; keep a backup of the data folder and the installer.	Avoids last-minute breakage.

10. Appendix: configuration of the delivered system

Item	Value
Platform	Windows WPF application (.NET Framework 4.7.2), CPU inference with ONNX Runtime; touchscreen mini PC, webcam, QR701 58 mm thermal printer

Item	Value
Conditions	Acne, hyperpigmentation, eczema (multi-label); normal skin derived
Decision limits (calibrated probability)	Acne 0.42, hyperpigmentation 0.37, eczema 0.27; margin 0.0
Lesion detector thresholds	Acne 0.275, hyperpigmentation 0.225, eczema 0.25; each condition's detector is an ensemble of three models
Certainty rule	Detected: high at probability 0.80 or more, moderate 0.60 or more, otherwise low. Not detected: high at 0.10 or less, moderate 0.25 or less. Detector-led finding: moderate if best box confidence is 0.50 or more, otherwise low
Unusual-photograph check	Embedding distance above 14.74 (99th percentile of training photographs)
Reading time	About 2 s per photograph on CPU; live reading every 2-3 s
Data folder	LUMYVUE\Patient Records\ (by verification status), lumyvue_db.sqlite, Backups\
Automated tests	103 passing